

Experiment 4 — FPGA-Based Traffic Monitoring & Signal Control System

RTL module: Traffic_Monitoring Testbench: tb_Traffic_Monitoring Simulation duration: 80 ns

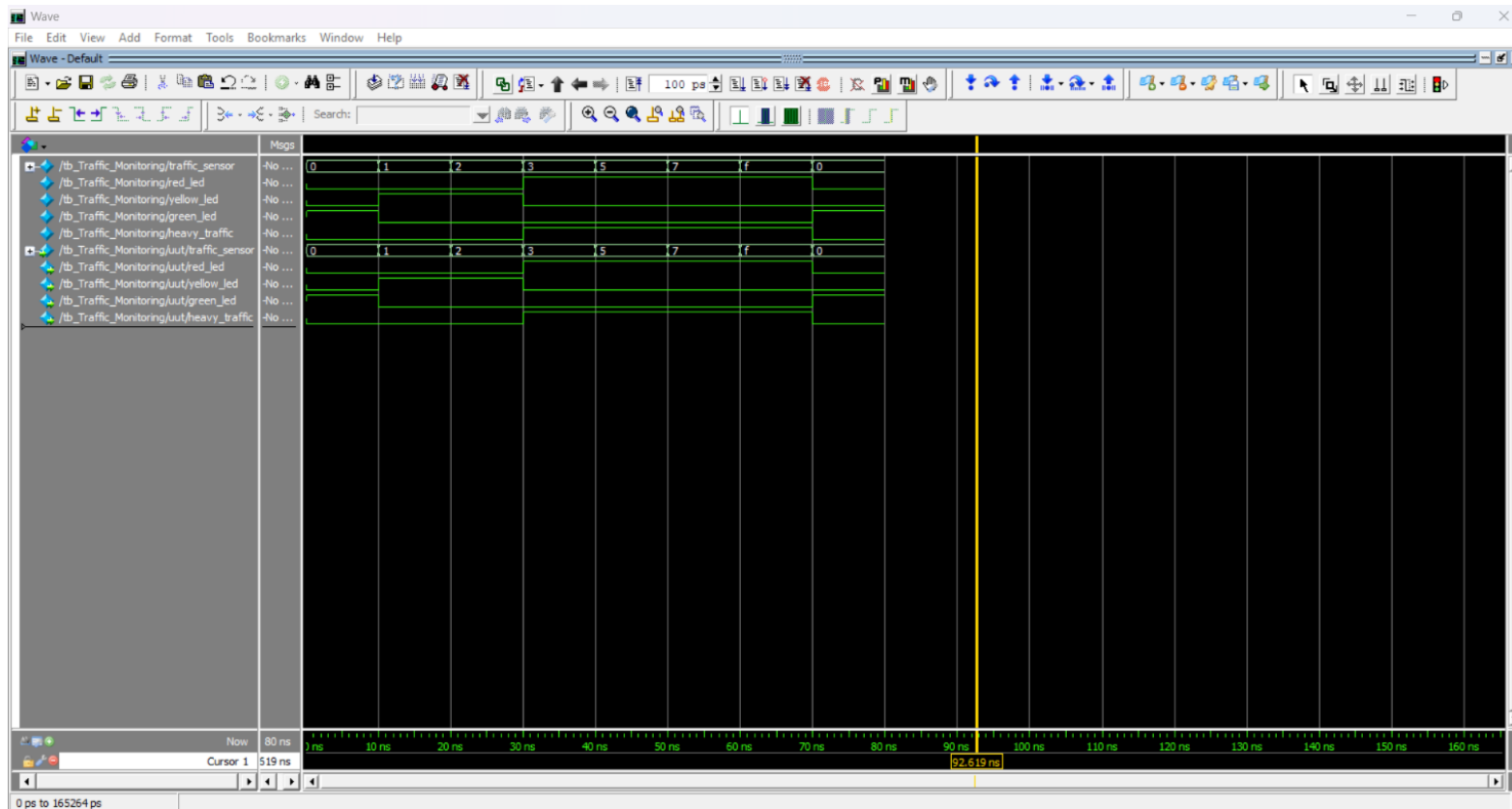
Simulation Objective

The simulation verifies traffic-state classification. No active sensors produce green, one active sensor produces yellow, and two or more active sensors produce red with the heavy-traffic indication.

Verification Sequence

Time	Input	Condition	Out-1	Out-2	Out-3	Out-4
0–10 ns	0000	No traffic	0	0	1	0
10–20 ns	0001	Light traffic	0	1	0	0
20–30 ns	0010	Light traffic	0	1	0	0
30–40 ns	0011	Heavy traffic	1	0	0	1
40–50 ns	0101	Heavy traffic	1	0	0	1
50–60 ns	0111	Heavy traffic	1	0	0	1
60–70 ns	1111	Heavy traffic	1	0	0	1
70–80 ns	0000	No traffic	0	0	1	0

Waveform Evidence — The screenshot below is the captured Questa waveform for this experiment.



Result: The waveform completes the 80 ns test sequence and demonstrates the expected logical transitions for traffic_sensor, red_led, yellow_led, green_led, heavy_traffic. The simulation stops normally at the end of the testbench.

Conclusion

Simulation verification completed successfully with the DUT and corresponding testbench loaded in Questa. The observed transitions are consistent with the intended RTL logic.